

CONTEXT-FREE GRAMMAR:

A CFG is a 4 tuple,

$$G = (V, T, S, P)$$

$\downarrow$
 Σ

where V and T/Σ are disjoint finite sets,

$S \in V \rightarrow$ starting production, and P is a finite set of formulas/productions of the form $(A \rightarrow \alpha)$, where $A \in V$ and $\alpha \in (V \cup \Sigma)^+$

- elements of T/Σ are called terminal symbols
- elements of V are called variables / non-terminals
- S is the start symbol
- elements of P are grammar rules or production

→ Context Free Language:

the language generated by CFG.

-if $G = (V, T, S, P)$ is a CFG, the language generated by G is

$$L(G) = \{ x \in \Sigma^+ \mid S \xRightarrow{G}^+ x \}$$

- a language L is a context-free language (CFL) if there is a CFG, G , with $L = L(G)$

Q1. Write a grammar and its rules to generate even length strings

Soln $L = \{x \mid |x| \text{ is even}\}$ $\Sigma = \{a, b\}$

$$L = \{\epsilon, aa, ab, ba, bb, \dots\}$$

$$G = (V, T, S, P)$$

$$\Sigma = \{a, b\}$$

$$V = \{$$

$$S = S$$

$$P = \{S \rightarrow \epsilon$$

$$S \rightarrow aSb \mid aSa \mid bSa \mid bSb$$

}

1. ϵ belongs to the language

2. $ab \Rightarrow aSb$,

(2) aSa ,

$$bSa$$

$$bSb$$

Q2. Write a CFG to generate the strings of the language L
 $L = \{w \mid |w| \text{ is odd and middle symbol is 'a'}\}$

Soln $L = \{a, aaa, aab, aba, \dots\}$

Der

$$S \rightarrow a$$

$$S \rightarrow aSa \mid aSb \mid bSa \mid bSb$$

Q3. Write a CFG.

$L = \{w \mid \text{first and middle symbol are same, and } |w| \text{ is odd}\}.$

Soln

$$L = \{ \quad aab, bba, bbb, aaa, \dots \}$$

$$V = \{ \quad \}$$

$$T = Z = \{a, b\}$$

$$S = S$$

$$S \rightarrow aAa \mid bBa \mid bBb$$

$$A \rightarrow a \mid aAa \mid aAb \mid bAa \mid bAb$$

$$B \rightarrow b \mid aBa \mid aBb \mid bBa \mid bBb$$

Q4 $L = \{w \mid |w| \text{ is odd, middle and last symbol are same}\}$

Soln

$$S \rightarrow aAa \mid aBb \mid bAa \mid bBb$$

$$A \rightarrow a \mid aAa \mid aAb \mid bAa \mid bAb$$

$$B \rightarrow b \mid aBa \mid aBb \mid bBa \mid bBb$$

Q5. $L = \{w \mid |w| \text{ is odd, first, middle and last symbol are same}\}$
Soln

$S \rightarrow aAa \mid bBb$

$A \rightarrow a \mid aAa \mid aAb \mid bAb \mid bAa$

$B \rightarrow b \mid aBa \mid aBb \mid bBb \mid bBa$

Q6. $L = \{a^n b^n \mid n \geq 0\}$

Soln $L = \{ab, aabb, \dots\}$

$S \rightarrow aSb \mid ab$

Q7. $L = \{w \mid |w| \text{ is odd}\} \rightarrow \text{CFG}$

Soln ① a, b are in the language - Base case

② If $w \in L$, then so is awa, awb, bwa, bwb are in the language and others are not.

$S \rightarrow a \mid b \mid aSa \mid aSb \mid bSa \mid bSb$

Q8. Write CFG for CFL $L = \{a^n b^n \mid n \geq 0\}$

Soln

① ϵ belongs to the language

② if $w \in L$, then so is $awb \in L$ and others do not.

$S \rightarrow \epsilon$

$S \rightarrow aSb$

Q9. Write CFG for CFL $L = \{a^n b^n \mid n > 0\}$

Soln $L = \{ab, aabb, \dots\}$

① $ab \in L$

② if $w \in L$, then so is $awb \in L$ others not

$S \rightarrow ab$

$S \rightarrow asb$

Q10. Write CFG for CFL $L = \{a^n b^n \mid n \geq 2\}$

Soln $L = \{aabb, aaabbb, \dots\}$

① $aabb \in L$

② if $w \in L$, then so is $awb \in L$, others not

$S \rightarrow aabb$

$S \rightarrow asb$

Q11. Write CFG for $L = \{a^n b^m c^k \mid n, m, k \geq 0, n = m+k\}$

Soln

$L = \{\epsilon, ab, ac, \dots\}$

$$n = m+k$$

$$\begin{aligned} a^n b^m c^k &= a^{m+k} b^m c^k \\ &= \underbrace{a^k a^m}_{\text{}} b^m c^k \end{aligned}$$

$$\begin{aligned} S &\rightarrow a S c \mid \epsilon \mid S_1 \\ S_1 &\rightarrow a S_1 b \mid a b \end{aligned}$$

Q12. CFG for language for $L = \{a^n b^m \mid n \geq m\}$

Soln $L = \{\epsilon, a, aa, \dots, ab, aab, aaab, \dots, aabb, aaabb, \dots\}$

$$\begin{aligned} S &\rightarrow \epsilon & a^{m+k} b^m & \text{when } n = m+k \\ S &\rightarrow a S b \mid A & a^k a^m b^m & k \geq 0 \\ A &\rightarrow a A \mid a \end{aligned}$$

Q13. Generate CFG $L = \{a^i b^j c^k \mid i, j, k \geq 0, j=i \text{ or } j=k\}$

Soln $j=i$
 $\underbrace{a^i b^j c^k} \rightarrow S_1$

$j=k$
 $\underbrace{a^i b^k c^k} \rightarrow S_2$

$$\begin{aligned} S_1 &\rightarrow a S_3 b C \mid C \\ C &\rightarrow c C \mid \epsilon \\ S_3 &\rightarrow a S_3 b \mid \epsilon \end{aligned}$$

$$\begin{aligned} S_2 &\rightarrow A b S_4 c \mid A \\ S_2 &\rightarrow A a \mid \epsilon \\ S_4 &\rightarrow b S_4 c \mid \epsilon \end{aligned}$$

⇒ Left Most Derivation (LMD) & Right Most Derivation (RMD):

- A derivation in a context free grammar (CFG) is a LMD if at each step a production is applied to the left most variable occurrence in the current string.
- A derivation in a CFG is a RMD if at each step a production is applied to the right most variable occurrence in the current string.

👉 Leftmost Derivation (LMD)

A leftmost derivation is a sequence of production steps where, at every step, the **leftmost non-terminal** symbol in the current sentential form is the one that is expanded (replaced) using a production rule. 📄

Feature	Description
Rule	Always expand the leftmost non-terminal symbol.
Notation	Often denoted with a subscript L : $\Rightarrow_L$ or $\xRightarrow{L}$
Significance	This order of derivation is typically generated by top-down parsers (like LL parsers).

📄 Export to Sheets



Example (LMD)

Consider the grammar for simple arithmetic expressions: $E \rightarrow E + T \mid T$ and $T \rightarrow a$. To derive the string $a + a$ using LMD:

$$\begin{aligned} E &\Rightarrow_L E + T && \text{(Expand leftmost } E \text{ first)} \\ &\Rightarrow_L T + T && \text{(Expand leftmost } T \text{ next)} \\ &\Rightarrow_L a + T \\ &\Rightarrow_L a + a \end{aligned}$$

👉 Rightmost Derivation (RMD)

A rightmost derivation is a sequence of production steps where, at every step, the **rightmost non-terminal** symbol in the current sentential form is the one that is expanded (replaced) using a production rule. 📄

Feature	Description
Rule	Always expand the rightmost non-terminal symbol.
Notation	Often denoted with a subscript R : $\Rightarrow_R$ or $\xRightarrow{R}$
Significance	This order of derivation is typically generated by bottom-up parsers (like LR parsers).

📄 Export to Sheets



Example (RMD)

Using the same grammar: $E \rightarrow E + T \mid T$ and $T \rightarrow a$. To derive the string $a + a$ using RMD:

$$\begin{aligned} E &\Rightarrow_R E + T && \text{(Expand rightmost } T \text{ first)} \\ &\Rightarrow_R E + a \\ &\Rightarrow_R T + a && \text{(Expand rightmost } T \text{ next)} \\ &\Rightarrow_R a + a \end{aligned}$$

🔑 Key Relationship to Parse Trees and Ambiguity

Both the LMD and RMD can be used to generate the same string, and both correspond to a single **Parse Tree** (Derivation Tree) for that string, **provided the grammar is unambiguous**. 🔄

- **Unambiguous Grammar:** A string has **exactly one** Parse Tree, which means it has **exactly one** LMD and **exactly one** RMD (though the LMD and RMD sequences are usually different, as shown in the example above).
- **Ambiguous Grammar:** A string has **more than one** Parse Tree. This means it will have **more than one** LMD and/or **more than one** RMD. Ambiguity is often formally defined by the existence of multiple LMDs (or RMDs) for the same string. 🔄

Would you like to see another example of LMD and RMD for an ambiguous grammar to highlight the multiple derivations?

⇒ Ambiguity in CFG:

A CFG is ambiguous if for at least one $x \in L(G)$, x has more than one derivation/parse tree or equivalently more than one LMD or equivalently one RMD.

Ambiguity in a **Context-Free Grammar (CFG)** occurs when a single terminal string in the language can be derived in **more than one distinct way**.

This structural uncertainty is critical in computer science, especially in **compiler design**, because a program's structure determines its meaning (semantics). If a piece of code (a string) has multiple valid structures (parse trees), it has multiple possible meanings, which is unacceptable for predictable execution.

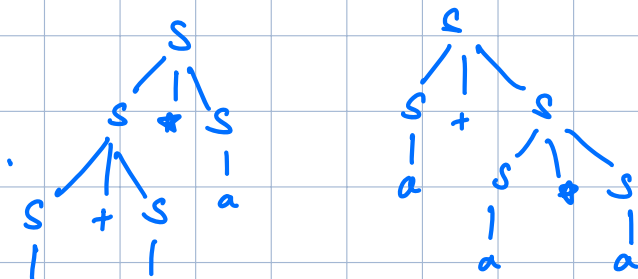
A CFG is called **ambiguous** if and only if there exists a string w in the language $L(G)$ such that w has:

1. **More than one Parse Tree (Derivation Tree).** (The defining characteristic).
2. **More than one Leftmost Derivation (LMD).**
3. **More than one Rightmost Derivation (RMD).**

Q14. Consider the CFG $S \rightarrow S + S \mid S * S \mid (S) \mid a$

$x = a + a * a$, $x \in L(G)$

Soln



a

a



2 LMDs.

To avoid ambiguity $\rightarrow$ we use a THUMB RULE

|

take highest priority precedence to the furthest level because compiler does from last

Example:

 $S \rightarrow aB|bA$
 $A \rightarrow a|aS|bAA$
 $B \rightarrow b|bS|aBB$

Find LMD, RMD, Parse Tree

String: bbaaa bba ba

Solⁿ

LMD:

 $S \rightarrow bA \quad (S \rightarrow bA)$
 $S \rightarrow bbAA \quad (A \rightarrow bAA)$
 $S \rightarrow bbaSA \quad (A \rightarrow aS)$
 $S \rightarrow bbaaBA \quad (S \rightarrow aB)$
 $S \rightarrow bbaaaBBA \quad (B \rightarrow aBB)$
 $S \rightarrow bbaaaBSBA \quad (B \rightarrow bS)$
 $S \rightarrow bbaaa bbaBA \quad (S \rightarrow bA)$
 $S \rightarrow bbaaa bbaBA \quad (A \rightarrow a)$
 $S \rightarrow bbaaa bba bA \quad (B \rightarrow b)$

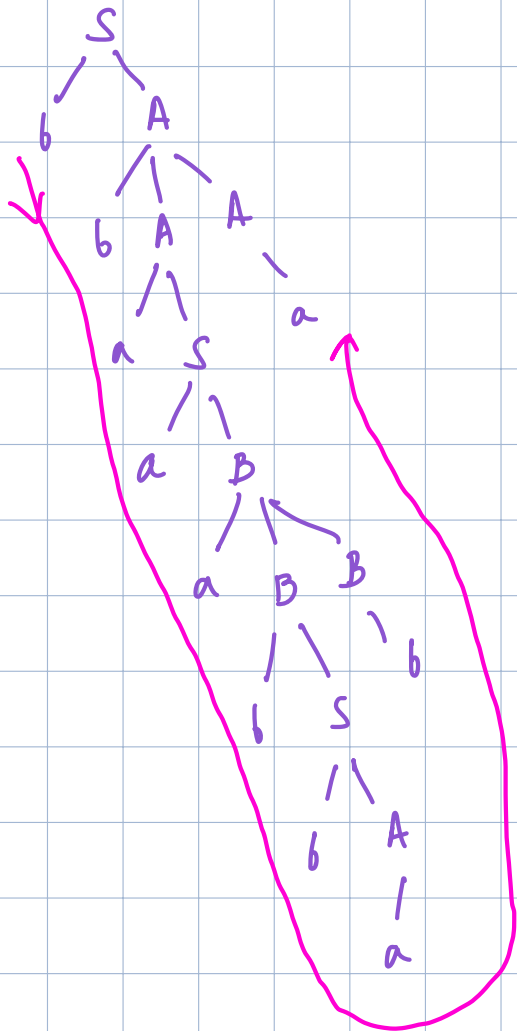
RMD

 $S \rightarrow bA \quad (S \rightarrow bA)$
 $S \rightarrow bbAA \quad (S \rightarrow bA)$
 $S \rightarrow bbaa \quad (A \rightarrow a)$
 $S \rightarrow bbaSa \quad (A \rightarrow aS)$
 $S \rightarrow bbaaBa \quad (S \rightarrow aB)$
 $S \rightarrow bbaaaBBa \quad (B \rightarrow aBB)$
 $S \rightarrow bbaaaBba \quad (B \rightarrow b)$
 $S \rightarrow bbaaaBSba \quad (B \rightarrow bS)$
 $S \rightarrow bbaaa bba ba \quad (S \rightarrow bA)$

$S \rightarrow bbaaaabbaba \ (A \rightarrow a)$

$S \rightarrow bbaaaabbaba \ (A \rightarrow a)$

Parse Tree:



example2:

$S \rightarrow 0B|1A$

$A \rightarrow 0|0S|1AA$

$B \rightarrow 1|1S|0BB$

String: '00110101'

Find LMD, RMD, Parse Tree

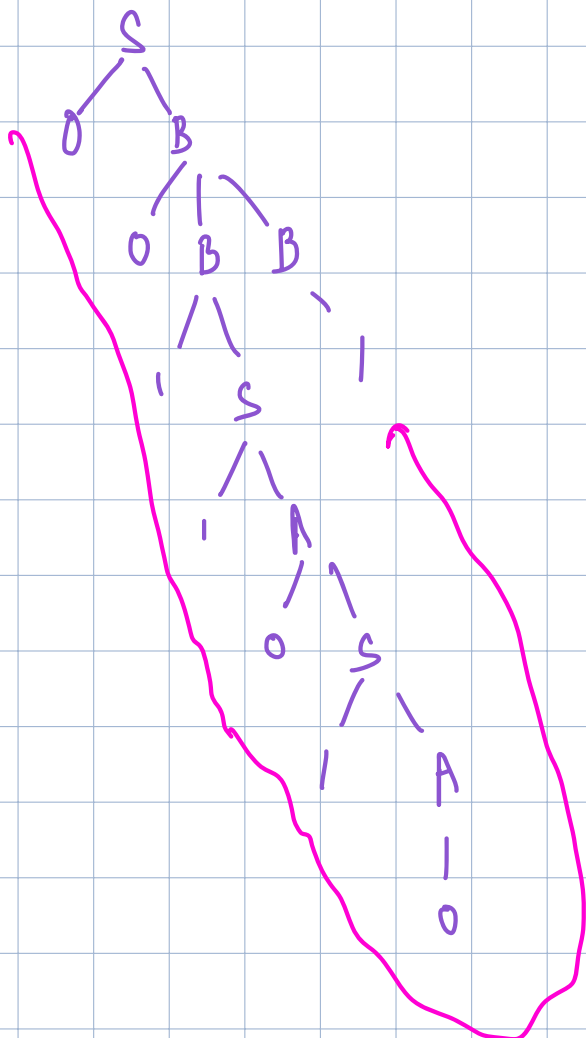
LMD

$S \rightarrow 0B (S \rightarrow 0B)$
 $S \rightarrow 00BB (B \rightarrow 0BB)$
 $S \rightarrow 001SB (B \rightarrow 1S)$
 $S \rightarrow 0011AB (S \rightarrow 1A)$
 $S \rightarrow 00110SB (A \rightarrow 0S)$
 $S \rightarrow 001101AB (S \rightarrow 1A)$
 $S \rightarrow 0011010B (A \rightarrow 0)$
 $S \rightarrow 00110101 (B \rightarrow 1)$

RMD

$S \rightarrow 0B (S \rightarrow 0B)$
 $S \rightarrow 00BB (B \rightarrow 0BB)$
 $S \rightarrow 00B1 (B \rightarrow 1)$
 $S \rightarrow 001S1 (B \rightarrow 1S)$
 $S \rightarrow 0011A1 (S \rightarrow 1A)$
 $S \rightarrow 00110S1 (A \rightarrow 0S)$
 $S \rightarrow 001101A1 (S \rightarrow 1A)$
 $S \rightarrow 00110101 (A \rightarrow 0)$

Parse Tree



Note: Grammar which is both left recursive and right recursive is ambiguous

Q) Consider the following grammar
 $E \rightarrow E + E \mid E * E \mid id$ String: "id+id*id"
Find: i) LMD ii) RMD iii) Construct a parse tree iv) Check whether the grammar is ambiguous or not.

Soln

1) $E \rightarrow E + E$

2) $E \rightarrow E * E$

3) $E \rightarrow id$

LMD :

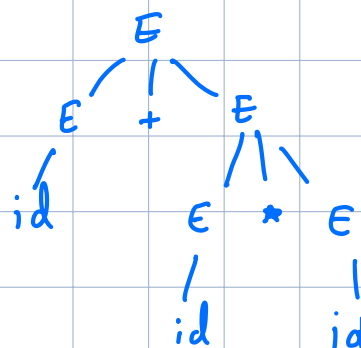
$E \rightarrow E + E$ ($E \rightarrow E + E$)

$E \rightarrow id + E$ ($E \rightarrow id$)

$E \rightarrow id + E * E$ ($E \rightarrow E * E$)

$E \rightarrow id + id * E$ ($E \rightarrow id$)

$E \rightarrow id + id * id$ ($E \rightarrow id$)



RMD:

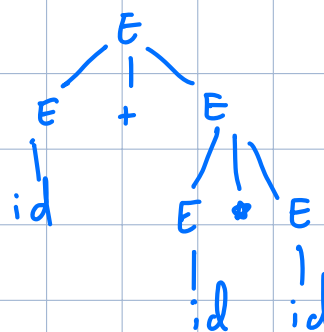
$E \rightarrow E + E$ ($E \rightarrow E + E$)

$E \rightarrow E + E * E$ ($E \rightarrow E * E$)

$E \rightarrow E + E * id$ ($E \rightarrow id$)

$E \rightarrow E + id * id$ ($E \rightarrow id$)

$E \rightarrow id + id * id$ ($E \rightarrow id$)



OR

$$E \rightarrow E * E \quad (E \rightarrow E * E)$$

$$E \rightarrow E + E * E \quad (E \rightarrow E + E)$$

$$E \rightarrow id + E * E \quad (E \rightarrow id)$$

$$E \rightarrow id + id * E \quad (E \rightarrow id)$$

$$E \rightarrow id + id * id \quad (E \rightarrow id)$$

since we got more than 1 LMD,
Grammar is ambiguous

⇒ Applications of CFGs:

(i) Describing programming languages

↳ there is a mechanical way for turning a CFG into a parser.

(ii) Describing DTD's in XML's :

- DTD: document type definition
- XML: extensible markUp language

(iii) NLP

(iv) Pattern Matching and Search

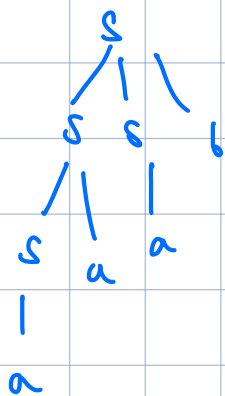
Q15. Show that the following grammar is ambiguous

$$S \rightarrow a | Sa | bSS | SSb | SbS$$

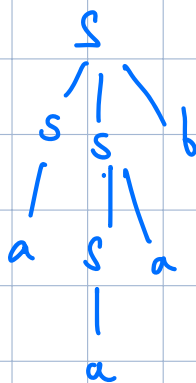
Solⁿ: $L = \{a, aa, aba, aaab \dots\}$

consider $aaab$

LMD1



LMD2



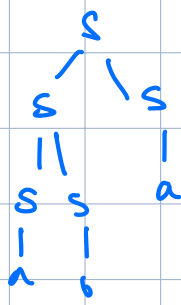
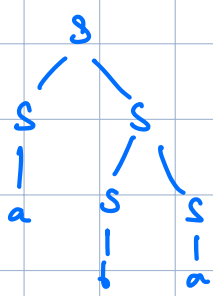
2 LMDs $\Rightarrow$ ambiguous

Q16. Show that the grammar is ambiguous

$$S \rightarrow SS | a | b$$

Solⁿ $L = \{a, b, ab, bba, abba, \dots\}$

consider $abba$



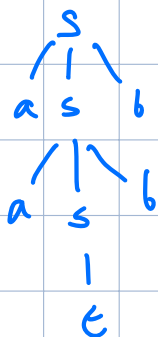
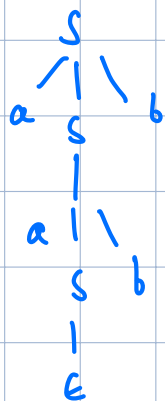
Q17. Show that the following grammar is not equivalent to an unambiguous grammar.

$$S \rightarrow aSb \mid aaSb \mid \epsilon$$

Solⁿ: $L = \{\epsilon, ab, aab, aaaSb, aaaSb, aaaSb, \dots\}$

consider

aabb



Unambiguous grammar:

$$S \rightarrow aSb \mid S_1$$

$$S_1 \rightarrow aaS_1b \mid \epsilon$$

Q18. Give CFG for:

(a) $L = a^n b^n$, n is even

Solⁿ $L = \{\epsilon, aabb, aaaaabbb, \dots\}$

$$S \rightarrow aaSbb \mid \epsilon$$

(b) $L = a^n b^n$, n is odd

Soln $L = \{ab, aaabbb, \dots\}$
 $S \rightarrow ab \mid aaSbb$

(c) $L = a^n b^n$, n is a multiple of 3

Soln $L = \{\epsilon, aaabbb, \dots\}$

$$n = 3m$$

$$L = a^{3m} b^{3m} \\ = (a^m b^m)^3$$

$$S \rightarrow \epsilon \mid aaaSbbb$$

(d) $L = a^n b^n$, n is not a multiple of 3.

Soln $L = \{ab, aabb, aaabbbb, \dots\}$

$$S \rightarrow ab \mid aSb$$

$$S \rightarrow aSb \mid ab$$

Q19. Give CFG for:

$$L \rightarrow \{a^n b^m : n \leq m+3 \mid n \geq 0, m > 0\}$$

$$L = \{\epsilon, ab, aab, aaab, abb, aabb, aaabb, \dots\}$$

$$S \rightarrow ABC$$

$$C \rightarrow bC \mid \epsilon$$

$$A \rightarrow aAa \mid aA \mid \epsilon$$

$$B \rightarrow aBb \mid \epsilon$$

Q20. Write CFG for $L = \{w \mid |w| \text{ is divisible by } 3\}$ $|w|$ is multiple of 3

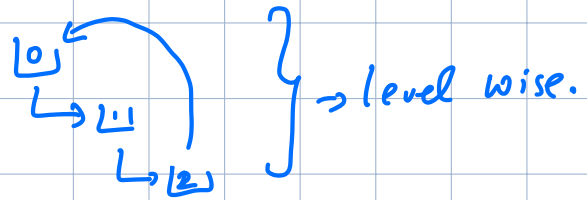
Solⁿ

$$L = \{\epsilon, aaa, aab, \dots\}$$

$$S \rightarrow aS_1 \mid bS_1 \mid \epsilon$$

$$S_1 \rightarrow aS_2 \mid bS_2$$

$$S_2 \rightarrow aS \mid bS$$



Q21. Write CFG for $L = \{w \mid |w| \text{ is ^{not} divisible by } 3\}$

Solⁿ

$$S \rightarrow aS_1 \mid bS_1$$

$$S_1 \rightarrow aS_2 \mid bS_2 \mid \epsilon$$

$$S_2 \rightarrow aS \mid bS \mid \epsilon$$

Q22. CFG for $L = \{w \mid |w| \% 2 = 0\}$

Solⁿ

$$S \rightarrow aS_1 \mid bS_1 \mid \epsilon$$

$$S_1 \rightarrow aS \mid bS$$

SIMPLIFICATION OF GRAMMAR:

- S1) Null Production Removal
- S2) Unit Production Removal
- S3) Useless Production/Variable Removal
- S4) Left Recursion Removal

⇒ Normal Forms:

① Chomsky Normal Form: (CNF)

- A context free grammar is said to be chomsky normal form if every production is one of these 2 types:

- (i) $A \rightarrow BC$ (where B and C are variables)
- (ii) $A \rightarrow \alpha$ (where α is a terminal symbol)

- to arrive at CNF, S1 and S2 are mandatory

- S3 is not mandatory, but is preferable as it reduces number of productions

→ Elimination of Null Productions :

- A production of the form $A \rightarrow \epsilon$ is known as null production or ϵ production

- Nullable variable:

A variable V, s is said to be nullable variable if it derives ϵ in 0 or more steps

ie. $S \xRightarrow{*} \epsilon$

eg: $S \rightarrow aA$
 $A \rightarrow aA | \epsilon$
 $\downarrow$
 NV

$S \rightarrow A$
 $A \rightarrow aA | \epsilon$
 $\downarrow$
 NV

Q23. Remove Null productions from G such that

$$S \rightarrow ABA$$

$$A \rightarrow aA | B$$

$$B \rightarrow bB | \epsilon$$

Solⁿ $NV = \{S, A, B\}$

$$S \rightarrow ABA | AA | B$$

$$A \rightarrow aA | B | a$$

$$B \rightarrow bB | b$$

Q24. Find NV of

$G: S \rightarrow ABA$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid \epsilon$

Solⁿ

$NV = \{B\}$

$S \rightarrow ABA$

$B \rightarrow b$

$A \rightarrow aA \mid a$

- to find the equivalent grammar without null production, steps are:

① Identify nullable variables

② Copy all the productions of G into G' , except null productions

③ Remove Nullable variables from the body of each production and introduce new production.

eg: $G: S \rightarrow AB \mid \epsilon$
 $A \rightarrow aASb \mid a$
 $B \rightarrow bS$

$NV = \{S\}$

$G' :$

$S \rightarrow AB$

$A \rightarrow aASb \mid a \mid aAb$

$B \rightarrow bS \mid b$

Q25. Remove nullable var for G :

$S \rightarrow AB$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow ab \mid bB \mid \epsilon$

Solⁿ $NV = \{S, A, B\}$

G' :

$S \rightarrow AB \mid B \mid A$

$A \rightarrow aA \mid a$

$B \rightarrow ab \mid bB \mid b$

Q26. Remove NV from G :

$S \rightarrow ABC \mid B a B$

$A \rightarrow aA \mid B a C \mid a a a$

$B \rightarrow b B b \mid a$

$C \rightarrow C A \mid A C$

Solⁿ No NV

Q27. Remove NV for G :

$S \rightarrow AB \mid ABC$

$A \rightarrow BA \mid BC \mid a \mid \epsilon$

$B \rightarrow AC \mid CB \mid b \mid \epsilon$

$C \rightarrow BC \mid AB \mid A \mid c$

Solⁿ $NV = \{S, A, B, C\}$

G' :

$S \rightarrow AB|ABC|B|A|BC|AC|C$

$A \rightarrow AB|BC|a|B|A|C$

$B \rightarrow AC|CB|b|A|C|B$

$C \rightarrow BC|AB|A|c|C|B$

→ Unit Production :

- A unit production is of the form $A \rightarrow B$, where A and B are variables
- unit productions are removed by substitution

eg: G :

$S \rightarrow aS|B|bB$

$B \rightarrow bB|b$

$\Downarrow$

G' : $S \rightarrow aS|bB|b$

$B \rightarrow bB|b$

Note: $S \rightarrow b$ ^{terminal} is not unit production
 $S \rightarrow B$ is unit production
 $\downarrow$
 variable

Q28. Remove null and unit productions from G:

$$S \rightarrow AB$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

Soln

$$(i) \text{ NV} = \{S, A, B\}$$

$$G' : S \rightarrow AB \mid A \mid B$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$(ii) G' : S \rightarrow AB \mid aA \mid a \mid bB \mid b$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

Q29. Convert to CNF :

$$S \rightarrow AB \mid aA \mid a \mid bB \mid b$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

Soln

G:

$$S \rightarrow AB \mid a \mid b \mid x_a A \mid x_b B$$

$$x_a \rightarrow a$$

$$x_b \rightarrow b$$

$$A \rightarrow X_a A | a$$

$$B \rightarrow X_b B | b$$

Q 30. Find equivalent CFG with No useless variables./production

$$G: S \rightarrow ABC \mid BaB$$

$$A \rightarrow aA \mid BaC \mid aaa$$

$$B \rightarrow bBb \mid a$$

$$C \rightarrow CA \mid AC$$

Soln - C is the useless variable as it is non generating, even though it is reachable

- hence the equivalent CFG is:

$$G': S \rightarrow BaB$$

$$A \rightarrow aA \mid aaa$$

$$B \rightarrow bBb \mid a$$

} Remove all instances consisting useless variable.

- Now A is not reachable, so it is useless variable.

hence final grammar:

$$G'': S \rightarrow BaB$$

$$B \rightarrow bBb \mid a$$

Q31. Find equivalent CFG with no useless variables

G: $S \rightarrow AB \mid AC$
 $A \rightarrow aAb \mid bAa \mid a$
 $B \rightarrow bbA \mid aAB \mid AB$
 $C \rightarrow aCa \mid aDb$
 $D \rightarrow bD \mid aC$

Solⁿ

C, D are useless vars
- all productions are reachable but C, D are in a loop (non-generating)
G':

$S \rightarrow AB$
 $A \rightarrow aAb \mid bAa \mid a$
 $B \rightarrow bbA \mid aAB \mid AB$

Q32. Eliminate all useless productions.

$S \rightarrow aS \mid AB$
 $A \rightarrow bA$
 $B \rightarrow AA$

Solⁿ all are non-generating (but reachable) hence useless.

$L(G) = \{ \}$

Q33. Convert grammar to CNF.

Q: $S \rightarrow a | aA | B | C$

$A \rightarrow aB | \epsilon$

$B \rightarrow Aa$

$C \rightarrow cCD$

$D \rightarrow ddd$

Soln

①

- C is reachable but non-generating hence remove C
- if we remove C , D becomes unreachable, hence useless production

G' : $S \rightarrow a | aA | B$

$A \rightarrow aB | \epsilon$

$B \rightarrow Aa$

② Remove nullable var

$NV = \{A\}$

$S \rightarrow a | aA | B$

$A \rightarrow aB | a$

$B \rightarrow Aa$

③ Remove unit production

$S \rightarrow a | aA | Aa$

$A \rightarrow aB | a$

$B \rightarrow Aa$

④ Convert to CNF

$$S \rightarrow a \mid x_a A \mid A x_a$$

$$A \rightarrow x_a B \mid a$$

$$B \rightarrow A x_a$$

$$x_a \rightarrow a$$

Q3. Convert the following grammar to CNF

$$G: S \rightarrow aA \mid aBB$$

$$A \rightarrow aaA \mid \epsilon$$

$$B \rightarrow bB \mid bbC$$

$$C \rightarrow B$$

Solⁿ ① - here B, C are reachable but non-generating, hence useless

$$G': S \rightarrow aA$$

$$A \rightarrow aaA \mid \epsilon$$

② Remove NV

$$NV = \{A\}$$

$$G'': S \rightarrow aA$$

$$A \rightarrow aaA \mid aa$$

③ Remove Unit prod

$$G'' : S \rightarrow aA \\ A \rightarrow aaA | aa$$

④ Convert to CNF

$$G''' : S \rightarrow x_a A \\ A \rightarrow x_a T_1 \mid x_a x_a \\ x_a \rightarrow a \\ T_1 \rightarrow x_a A$$

Q35. Find the CFG with no ϵ productions, that generate the same language, except possibly for ϵ .

(i) $S \rightarrow AB \mid \epsilon$
 $A \rightarrow aASb \mid a$
 $B \rightarrow bS$

Soln $NV = \{S\}$

$G' :$

$$S \rightarrow AB$$

$$A \rightarrow aASb \mid aAb$$

$$B \rightarrow bS \mid b$$

$$L(G') = L(G) - \{\epsilon\}$$

(ii) $G: S \rightarrow AB|ABC$

$A \rightarrow BA|BC| \epsilon | a$

$B \rightarrow AC|CB| \epsilon | b$

$C \rightarrow BC|AB|A|c$

Solⁿ $NV: \{S, A, B, C\}$

$G': S \rightarrow AB|ABC|A(B)BC|AC|C$

$A \rightarrow BA|BC|a|B|A|C$

$B \rightarrow AC|CB|b|A|C|B$

$C \rightarrow BC|AB|A|c|B|C$

Q36. Find CFG G' with no ϵ productions and no unit productions that generate the language $L(G) - \{\epsilon\}$

(i) $G: S \rightarrow aSa|bSb| \epsilon$

$A \rightarrow aBb|bBa$

$B \rightarrow aB|bB| \epsilon$

Solⁿ A, B are useless, so eliminate them $\rightarrow$ unreachable.

$G: S \rightarrow aSa|bSb| \epsilon$

$\Downarrow$ removing $NV: \{S\}$

$G': S \rightarrow aSa|bSb|aa|bb$

no unit productions so stop

(ii) $G: S \rightarrow A|B|C$

$A \rightarrow aAa|B$

$B \rightarrow bB|bb$

$C \rightarrow aCaa|D$

$D \rightarrow baD|abD|aa$

Solⁿ No useless productions yet.

No nullable variables

↓ Removing Unit productions

$G' : S \rightarrow aAa|bB|bb|aCaa|baD|abD|aa$

$A \rightarrow aAa|bB|bb$

$B \rightarrow bB|bb$

$C \rightarrow aCaa|baD|abD|aa$

$D \rightarrow baD|abD|aa$

Q37. Given the CFF, find an equivalent CFG with no useless variables.

(i) $S \rightarrow ABC|BaB$

$A \rightarrow aA|BaC|aaa$

$B \rightarrow bBb|a$

$C \rightarrow CA|AC$

Solⁿ C is not generating $\rightarrow$ so useless

$$G': S \rightarrow BaB$$

$$B \rightarrow bBb \mid a$$

(ii)

$$S \rightarrow AB \mid AC$$

$$A \rightarrow aAb \mid bAa \mid a$$

$$B \rightarrow bbA \mid aaB \mid AB$$

$$C \rightarrow abCa \mid aDb$$

$$D \rightarrow bD \mid aC$$

Soln C, D are reachable but non-generating

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid bAa \mid a$$

$$B \rightarrow bbA \mid aaB \mid AB$$

Q38 Given the CFG, find a CFG G' in CNF, generating $L(G) - \{ \epsilon \}$.

(i) $S \rightarrow SS \mid (s) \mid \epsilon$

Soln (i) removing ϵ production ($NV = \{s\}$)

$$S \rightarrow SS \mid s \mid (s) \mid ()$$

→ just another character like 'a', 'b'

(ii) Eliminating unit production:

$$S \rightarrow SS \mid (S) \mid ()$$

(ii) Converting to CNF:

$$S \rightarrow SS \mid X_c T_1 \mid X_c X_2$$

$$T_1 \rightarrow SX_2$$

$$X_c \rightarrow ($$

$$X_2 \rightarrow)$$

GNF: $\rightarrow$ Peter TB.

$\hookrightarrow$ Greibach Normal Form

- for CFG to be in GNF, $[S_4, S_1, S_2]$ are mandatory, S_3 is optional but preferable
- a grammar $G = (V, T, P, S)$ is said to be in GNF, if all the productions, are of the form:

$$\boxed{A \rightarrow a\alpha}$$

where

$$\begin{aligned} A &\in V \\ a &\in T \\ \alpha &\in V^* \end{aligned}$$

Note: the body of the production, starts with a terminal followed by 0 or more variables

eg: $S \rightarrow aS \mid bSS \mid a$ ✓
 $D \rightarrow SS$ ✗

Q39. Convert the following to CNF form

(i) $S \rightarrow aSb \mid bSa \mid a \mid b$

Soln

GNF: G' :

$$S \rightarrow aS_x b \mid bS_x a \mid a \mid b$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

(ii) $S \rightarrow aSb \mid bSa \mid a \mid b \mid \epsilon$

Soln

$NV = \{S\}$

$$S \rightarrow aSb \mid ab \mid bSa \mid ba \mid a \mid b$$

GNF:

$$S \rightarrow aS_x b \mid aX_b \mid bS_x a \mid bX_a \mid a \mid b$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

⇒ Left Recursion: ★★★ (Important)

- a production of the form:

$$A \rightarrow A\alpha$$

where $A \in V$

$\alpha \in (V \cup T)^+$

is called left recursion

eg: Consider the unambiguous validation arithmetic expression grammar.

$$E \rightarrow E+T \mid E-T \mid T$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid a$$

Here:

$$E \rightarrow E+T$$

$$E \rightarrow E-T$$

$$T \rightarrow T * F$$

$$T \rightarrow T / F$$

are left recursion and others are not

→ Removal of Left Recursion:

Let $A \rightarrow A\alpha_1 \mid A\alpha_2 \mid \dots \mid A\alpha_m \mid \beta_1 \mid \beta_2 \mid \dots \mid \beta_n$,

└──────────┘└──────────┘

left recursionnon-left recursion.

then the equivalent grammar after removing left recursion is.

$$A \rightarrow \beta_1 A' \mid \beta_2 A' \mid \dots \mid \beta_n A'$$

$$A' \rightarrow \alpha_1 A' \mid \alpha_2 A' \mid \dots \mid \alpha_m A' \mid \epsilon$$

eg: $A \rightarrow Aa \mid b$

$$\alpha_1 = a$$

$$\beta_1 = b$$

$$A \rightarrow bA'$$

$$A' \rightarrow aA' \mid \epsilon$$

Q46. Find the non-left recursion:

$$E \rightarrow E+T \mid E-T \mid T$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid a$$

Soln

for. $E \rightarrow E+T \mid E-T \mid T$

$$\alpha_1 = +T$$

$$\alpha_2 = -T$$

$$\beta_1 = T$$

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid -TE' \mid \epsilon$$

$$\text{for: } T \rightarrow T * F \mid T / F \mid F$$

$$\alpha_1 = *F$$

$$\alpha_2 = /F$$

$$\beta_1 = F$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid /FT' \mid \epsilon$$

$$F \rightarrow (E) \mid a$$

G' :

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid -TE' \mid \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid /FT' \mid \epsilon$$

$$F \rightarrow (E) \mid a$$

(a) Eliminate

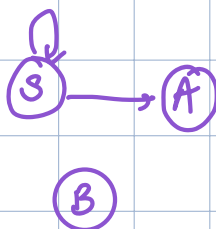
,

,

(

⇒ Dependency Graph:

eg : $S \rightarrow aSA$
 $A \rightarrow a$
 $B \rightarrow aa$



- they are a way of visualizing complex relationships and are found in many applications
- for CFGs, a dependency graph has its vertices labelled with variables, with an edge b/w the vertices, say, C and D iff there is a production of the form $C \rightarrow \alpha D \gamma$
- a variable is useful iff there is a path from the vertex say C to the vertex say D. In the above dependency graph, B is unreachable, hence B is useless.

Q41. Find the CFG with ϵ production equivalent to the given grammar G :

G : $S \rightarrow ABaC$

$A \rightarrow BC$

$B \rightarrow b| \epsilon$

$C \rightarrow D| \epsilon$

$D \rightarrow d$

Solⁿ

$$NV = \{A, B, C\}$$

all the variables are reachable and generating, so no useless variable at this step.

$$G': S \rightarrow ABaC \mid BaC \mid AaC \mid ABa \mid aC \mid Aa \mid Ba \mid a$$

$$A \rightarrow BC \mid B \mid C$$

$$B \rightarrow b$$

$$C \rightarrow D$$

$$D \rightarrow d$$

Q.42. For the given grammar remove all unit productions.

$$G: S \rightarrow Aa \mid B$$

$$B \rightarrow A \mid bb$$

$$A \rightarrow a \mid bc \mid B$$

Solⁿ

$$G': S \rightarrow Aa \mid a \mid bc \mid bb$$

$$B \rightarrow a \mid bc \mid bb$$

$$A \rightarrow a \mid bc \mid bb$$

Q.43. Show that the two grammars are equivalent

$$G_1: S \rightarrow abAB \mid ba$$

$$A \rightarrow aaa$$

$$B \rightarrow aA \mid bb$$

$$G_2: S \rightarrow abAaA \mid abA \mid bb \mid ba$$

$$A \rightarrow aaa$$

Solⁿ Replacing the production of B in G_1 we get S_2

Q44. Eliminate all useless production from the grammar

$$G: S \rightarrow aS \mid AB$$

$$A \rightarrow bA$$

$$B \rightarrow AA$$

What language does the grammar generate?

Solⁿ

All of the variables are useless, as all of them are reachable but not generating

$$L = \{ \}$$

Q45. Remove ϵ productions

$$S \rightarrow AaB \mid aaB$$

$$A \rightarrow \epsilon$$

$$B \rightarrow bbA \mid \epsilon$$

Solⁿ

$$NV = \{ A, B \}$$

$$S \rightarrow AaB \mid aB \mid Aa \mid a \mid aaB \mid aa$$

$$B \rightarrow bbA \mid bb$$

A is eliminated

$$S \rightarrow aB|a|aaB|aa$$
$$B \rightarrow bb$$

Q.46. Remove all unit productions, useless productions, and ϵ productions from

G: $S \rightarrow aA|aBB$

$$A \rightarrow aaA|\epsilon$$
$$B \rightarrow bB|bbc$$
$$C \rightarrow B$$

Solⁿ B, C are reachable but not generating $\rightarrow$ useless

or

$$S \rightarrow aA$$
$$A \rightarrow aaA|\epsilon$$

|| Remove ϵ tran.

$$NV = \{A\}$$
$$S \rightarrow aA|a$$
$$A \rightarrow aaA|aa$$

there is not unit production

Q.47. Convert the grammar to CNF.

$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$

Solⁿ

$$S \rightarrow AB|a \rightarrow \text{CNF}$$

$$S \rightarrow AT_1$$

$$T_1 \rightarrow BXa$$

$$Xa \rightarrow a$$

$$A \rightarrow XaT_2$$

$$T_2 \rightarrow XaXb$$

$$Xb \rightarrow b$$

$$B \rightarrow AXc$$

$$Xc \rightarrow c$$

Q43. Convert the grammar to GNF.

$$S \rightarrow AB$$

$$A \rightarrow aA | bB | b$$

$$B \rightarrow b$$

Solⁿ

$$S \rightarrow a\alpha$$

$$S \rightarrow aAB | bBB | bB$$

$$A \rightarrow aA | bB | b$$

$$B \rightarrow b$$

Q49. Convert to GNF

$$S \rightarrow abSb \mid aa$$

Soln

$$S \rightarrow aX_bSX_b \mid aX_a$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

Q50. $S \rightarrow AA \mid 0$

$$A \rightarrow SS \mid 1$$

Soln

$$S \rightarrow AA \mid 0$$

$$\alpha_1 = A$$

$$\beta_1 = 0$$

$$S \rightarrow 0A'$$

$$A' \rightarrow AA' \mid \epsilon$$

$$A \rightarrow SS \mid 1$$

$$\alpha_1 = S$$

$$\beta_1 = 1$$

$$A \rightarrow 1A''$$

$$A'' \rightarrow SA''$$

Assignment 2: 12/Nov/2025

1. Find context-free grammars for the following languages:

- (a) $L = a^n b^n$, n is even.
- (b) $L = a^n b^n$, n is odd.
- (c) $L = a^n b^n$, n is a multiple of three.
- (d) $L = a^n b^n$, n is not a multiple of three.

2. Consider the grammar $G = (\{S\}, \{a, b\}, S, P)$, with productions

$$S \rightarrow aSa \mid bSb \mid \lambda$$

Show a derivation tree for $w = aaabbaaa$ using the grammar

3. Consider the grammar G , with productions

$$\begin{aligned} S &\rightarrow abB, \\ A &\rightarrow aaBb, \\ B &\rightarrow bbAa, \\ A &\rightarrow \lambda, \end{aligned}$$

Give a derivation tree for $w = abbbaabbaba$ for the grammar.

4. Find context-free grammars for the following languages (with $n \geq 0$, $m \geq 0$).

- (a) $L = \{a^n b^m : n \leq m + 3\}$.
- (b) $L = \{a^n b^m : n = m - 1\}$.
- (c) $L = \{a^n b^m : n \neq 2m\}$.
- (d) $L = \{a^n b^m : 2n \leq m \leq 3n\}$.
- (e) $L = \{w \in \{a, b\}^* : na(w) \neq nb(w)\}$.
- (f) $L = \{w \in \{a, b\}^* : na(v) \geq nb(v), \text{ where } v \text{ is any prefix of } w\}$.
- (g) $L = \{w \in \{a, b\}^* : na(w) = 2nb(w) + 1\}$.
- (h) $L = \{w \in \{a, b\}^* : na(w) = nb(w) + 2\}$

NOTE: $na(w)$ means number of a's in the string w

5. What language does the grammar generate?

$$\begin{aligned} S &\rightarrow aSB \mid bSA \\ S &\rightarrow \lambda \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

6. What language does the grammar generate?

$$\begin{aligned} S &\rightarrow aaSbb \mid SS \\ S &\rightarrow \lambda \end{aligned}$$

7. Find context-free grammars for the following languages (with $n \geq 0$, $m \geq 0$, $k \geq 0$):

- (a) $L = \{a^n b^m c^k : n = m \text{ or } m \leq k\}$.
- (b) $L = \{a^n b^m c^k : n = m \text{ or } m \neq k\}$.
- (c) $L = \{a^n b^m c^k, k = n + m\}$.
- (d) $L = \{a^n b^m c^k, k = |n - m|\}$.
- (e) $L = \{a^n b^m c^k, k = n + 2m\}$.
- (f) $L = \{a^n b^m c^k, k \neq n + m\}$.

8. Find a context-free grammar for $L = \{w \in \{a, b, c\}^* \mid na(w) + nb(w) \neq nc(w)\}$.

Assignment Answers :

1. (a) $a^n b^n$, n is even

$$n = 2k$$

$$a^{2k} b^{2k} = (a^k b^k)^2 = (aa)^k (bb)^k$$

$$S \rightarrow aaSbb \mid \epsilon$$

(b) $a^n b^n$, n is odd

$$n = 2k + 1$$

$$a^{2k+1} b^{2k+1} = a a^{2k} b^{2k} b$$

$$S \rightarrow aS_1 b$$

$$S_1 \rightarrow aaS_1 bb \mid \epsilon$$

$$\text{or } S \rightarrow ab \mid aaS bb$$

(c) $a^n b^n$, $n = \text{multiple of } 3$

$$n = 3k$$

$$(aaa)^k (bbb)^k$$

$$S \rightarrow aaaS bbb \mid \epsilon$$

(d) $a^n b^n$, $n = \text{not multiple of 3}$

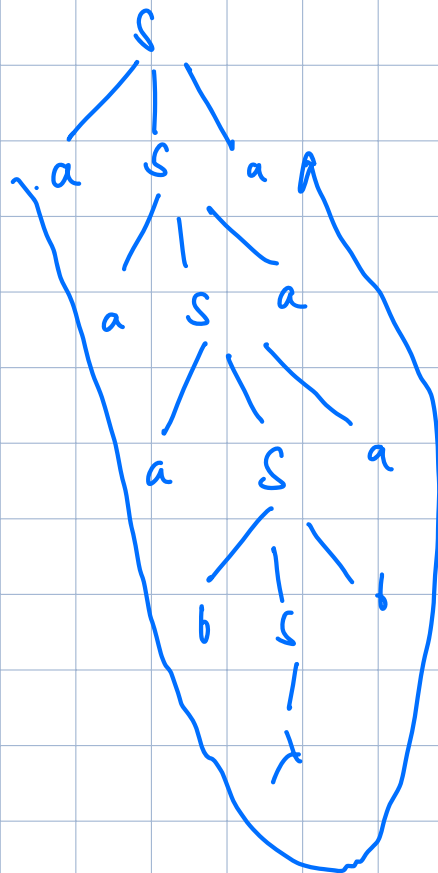
$$S \rightarrow ab \mid aS_1b$$

$$S \rightarrow aaaSbbb \mid ab \mid aabb$$

$$S_1 \rightarrow as_1b \mid ab$$

2. $S \rightarrow \underline{aSa} \mid \underline{bSb} \mid \lambda$

$w = \underline{aaabbaaa}$



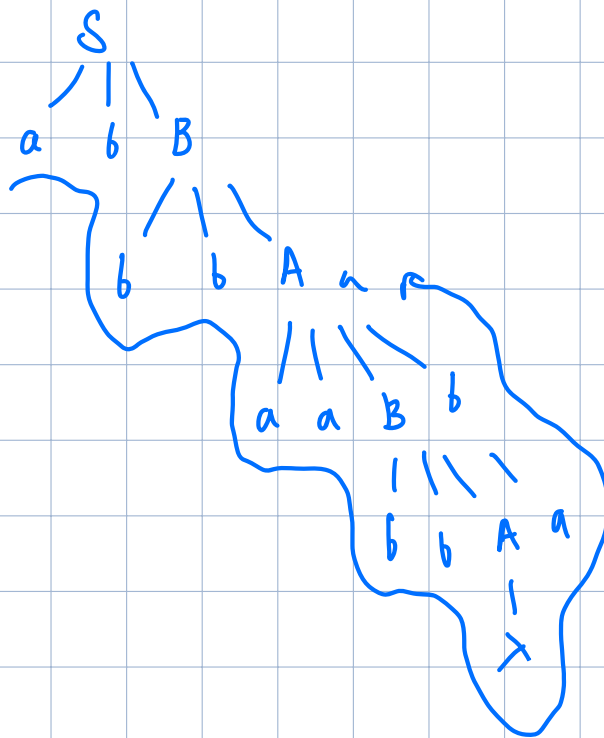
$$3. S \rightarrow abB$$

$$A \rightarrow \underline{a}aBb$$

$$B \rightarrow \underline{b}bAa$$

$$A \rightarrow \lambda$$

$$w = a \underline{b} b b a \underline{a} b b a b a$$



$$④ \quad n \geq 0, m \geq 0$$

$$(a) L = \{ a^n b^m : n \leq m+1 \}$$

$$L = \{ \underline{\epsilon}, a, aa, aaa, ab, aab, aaab, aaaaab, b, \dots \}$$

$$S \rightarrow aSb \mid T \mid \epsilon$$

$$T \rightarrow aT \mid aa \mid \epsilon$$

$$S \rightarrow ABC$$

$$C \rightarrow bC \mid \epsilon$$

$$A \rightarrow a \mid aa \mid aaa \mid \epsilon$$

$$B \rightarrow aBb \mid \epsilon$$

$$S \rightarrow AAAB \mid$$

$$A \rightarrow a \mid \epsilon$$

$$B \rightarrow aBb \mid Bb \mid \epsilon$$

$$S \rightarrow T \mid aT \mid aaT \mid aaaT$$

$$T \rightarrow aTb \mid B$$

$$B \rightarrow bB \mid \epsilon$$

$$S \rightarrow aSb \mid B \mid C$$

$$B \rightarrow Bb \mid \epsilon$$

$$C \rightarrow a \mid aa \mid aaa$$

(b) $L = \{a^n b^m : n = m - 1\}$

$$a^n b^{n+1}$$

$$a^n b^n b$$

$$S \rightarrow b \mid aSb$$

$$(c) L = \{ a^n b^m : n \neq 2m \}$$

$$n < 2m$$

$$S \rightarrow aS_1b / aBb / ab / B$$

$$B \rightarrow bB / b$$

$$S \rightarrow S_1 / S_2 / ab / a / b$$

$$n > 2m$$

$$S_2 \rightarrow aaS_2b / aaAb / A$$

$$A \rightarrow aA / a$$

$$L = \{ a, b, aabb, aaab, abb, \dots \}$$

$$aaaa bbb$$

$$S \rightarrow S_1 / S_2 / a / b / ab$$

$$S_1 \rightarrow aS_1b / aBb / B$$

$$B \rightarrow bB / b$$

$$S_2 \rightarrow aaS_2b / aaAb / A$$

$$A \rightarrow aA / a$$

$$(d) L = \{ a^n b^m : 2n \leq m \leq 3n \}$$

$$L = \{ \epsilon, ab, abbb, \underline{aaabbbb}, aabbbbbb, aabbbbbb, aabbbbbb, \dots \}$$

$$S \rightarrow \epsilon / aSbb / aSbbb$$

$$aaabbbbb$$

$$aaabbbbb$$

$$\underline{aaabbbbb}$$

$$aa \dots$$

$$(e) L = \{ w \in (a,b)^* : n_a(w) = n_b(w) \}$$

$$L = \{ a, b, abb, bba, \dots \}$$

$$\begin{aligned} S &\rightarrow S_1 S_2 \\ E &\rightarrow aEb \mid bEa \mid EE \mid \epsilon \\ S_1 &\rightarrow \epsilon a E \mid S_1 S_1 \\ S_2 &\rightarrow \epsilon b E \mid S_2 S_2 \end{aligned}$$

$$(f) L = \{ w \in (a,b)^* : n_a(v) \geq n_b(v), \text{ where } v \text{ is any prefix of } w \}$$

$$S \rightarrow SS \mid aSb \mid aS \mid \epsilon$$

$$(g) L = \{ w \in (a,b)^* : n_a(w) = 2 n_b(w) + 1 \}$$

$$\{ \underline{a}, \underline{aaab}, \underline{abaa}, \underline{baaa}, \dots \}$$

$$\begin{aligned} S &\rightarrow a \mid SA \mid AS \mid aA \mid Aa \\ A &\rightarrow \epsilon \mid AA \\ A &\rightarrow aAb \\ A &\rightarrow aAab \end{aligned}$$

$$A \rightarrow aAb a$$

$$A \rightarrow A a n b$$

$$A \rightarrow A a b a$$

$$A \rightarrow A b a a$$

$$A \rightarrow a b a A$$

$$A \rightarrow b a a A$$

$$A \rightarrow a b A a$$

$$A \rightarrow b a A a$$

$$P = \{ S \rightarrow a a F \mid F a \mid S F \mid F S \\ F \rightarrow \epsilon \mid a F a F b F \mid a F b F a F \mid b F a F a F \mid F F \}$$

$$(h) L = \{ w \in (a,b)^* : n_a(w) = n_b(w) + 2 \}$$

$$L = \{ aa, aaab, baaa, abaa, aaba, \dots \}$$

$$S \rightarrow aa \mid S_1 a S_1 a S_1$$

$$S_1 \rightarrow \epsilon \mid a S_1 b S_1 \mid b S_1 a S_1 \mid S_1 S_1$$

$$(5) S \rightarrow a S B \mid b S A$$

$$S \rightarrow \lambda$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$L = \{ w(w^r) \}$$

complement

reverse

$$(6) S \rightarrow aaSbb \mid \epsilon \mid \lambda$$

Formal description of the language

$L = \{ w \in \{a, b\}^* \mid w = h(x) \text{ where } x \text{ is a Dyck word and } h^{(n)}(x) = aa, h^{(n)}(x) = bb \}$,

i.e., the set of all strings of the form:

$$w = (aa)^n \text{ arranged in balanced matched structures with } (bb)^n$$

where each matched pair has:

- two 'a's for each opening,
 - two 'b's for each closing,
- and the nesting structure is properly balanced.

$$(8) L = \{ w \in \{a, b, c\}^* \mid n_a(w) + n_b(w) = n_c(w) \}$$

$$S \rightarrow G_1 \mid G_2$$

$$E \rightarrow \epsilon \mid aEc \mid cEa \mid bEc \mid cEb$$

$$G_1 \rightarrow EaE \mid EbE \mid G_1G_1$$

$$G_2 \rightarrow EcE \mid G_2G_2$$

$$(7) n \geq 0, m \geq 0, k \geq 0$$

$$(a) L = \{ a^n b^m c^k : n=m \text{ or } m \leq k \}$$

$$S \rightarrow S_1 \mid S_2$$

$$\begin{matrix} m < k \\ m = k \end{matrix}$$

$$S_1 \rightarrow aS_3b \mid \epsilon$$

$$S_3 \rightarrow aS_3b \mid \epsilon$$

$$C \rightarrow c \mid \epsilon$$

$$S_2 \rightarrow AS_4$$

$$S_4 \rightarrow bS_4c \mid \epsilon$$

$$A \rightarrow aA \mid \epsilon$$

$$C \rightarrow Cc | \epsilon$$

$$S \rightarrow S_1 | S_2 | \epsilon$$

$$S_1 \rightarrow aS_3bC | \epsilon$$

$$S_2 \rightarrow AS_4$$

$$S_3 \rightarrow aS_3b | \epsilon$$

$$S_4 \rightarrow bS_4C | \epsilon$$

$$A \rightarrow aA | \epsilon$$

$$C \rightarrow Cc | \epsilon$$

$$\begin{aligned} S &\rightarrow AN | BM \\ A &\rightarrow aAb | \epsilon \\ B &\rightarrow aB | \epsilon \\ N &\rightarrow Nc | \epsilon \\ M &\rightarrow bMc | N \end{aligned}$$

$$(b) L = \{ a^n b^m c^k, n=m \text{ or } m! = k \}$$

$$S \rightarrow AM | BN | BN'$$

$$A \rightarrow aAb | \epsilon$$

$$M \rightarrow cM | c$$

$$B \rightarrow aB | \epsilon$$

$$N \rightarrow bNc | M$$

$$N' \rightarrow bB | b$$

$$(c) L = \{ a^n b^m c^k, k = n + m \}$$

$$\begin{aligned} a^n b^m c^n c^m \\ a^n (b^m c^m) c^n \end{aligned}$$

$$S \rightarrow S_1 | aSc$$

$$S_1 \rightarrow bS_1c | \epsilon$$

$$(d) L = \{a^n b^m c^k : k = |n-m|\}$$

$$k = n - m$$

$$n = k + m$$

$$k = m - n$$

$$m = n + k$$

$$a^{k+m} b^m c^k$$

$$a^k a^m b^m c^k$$

↓

$$S_1 \rightarrow S_2 | a S_1 c$$

$$S_2 \rightarrow b S_2 c | \epsilon$$

$$\underline{a^n b^n} \underline{b^k c^k}$$

$$S \rightarrow S_3 S_4$$

$$S_3 \rightarrow a S_3 b | \epsilon$$

$$S_4 \rightarrow b S_4 c | \epsilon$$

$$S \rightarrow S_1 | S_2$$

$$S_1 \rightarrow S_2 | a S_1 c$$

$$S_2 \rightarrow b S_2 c | \epsilon$$

$$S_2 \rightarrow S_4 S_5$$

$$S_4 \rightarrow a S_4 b | \epsilon$$

$$S_5 \rightarrow b S_5 c | \epsilon$$

$$(e) L = \{a^n b^m c^k : k = n + 2m\}$$

$$a^n b^m c^n c^{2m}$$

$$a^n \underline{b^m c^{2m}} c^n$$

$$S \rightarrow S_1 | a S c$$

$$S_1 \rightarrow b S_1 c c | \epsilon$$

$$(f) L = \{a^n b^m c^k \mid k! = n+m\}$$

$$S \rightarrow G \mid L$$

$$E \rightarrow E_1 \mid aEC$$

$$E_1 \rightarrow bE_1C \mid E$$

$$G \rightarrow aG \mid Eb$$

$$L \rightarrow$$

$$S \rightarrow aAS_1 \mid AS_2$$

$$S_1 \rightarrow aS_1C \mid bBT$$

$$S_2 \rightarrow aS_2C \mid BT$$

$$A \rightarrow aA \mid E$$

$$B \rightarrow bB \mid E$$

$$T \rightarrow bTc \mid E$$

$$S \rightarrow G \mid L_1 \mid L_2$$

$$E \rightarrow aEc \mid E_1$$

$$E_1 \rightarrow bE_1c \mid \lambda$$

$$G \rightarrow EC$$

$$C \rightarrow cC \mid \lambda$$

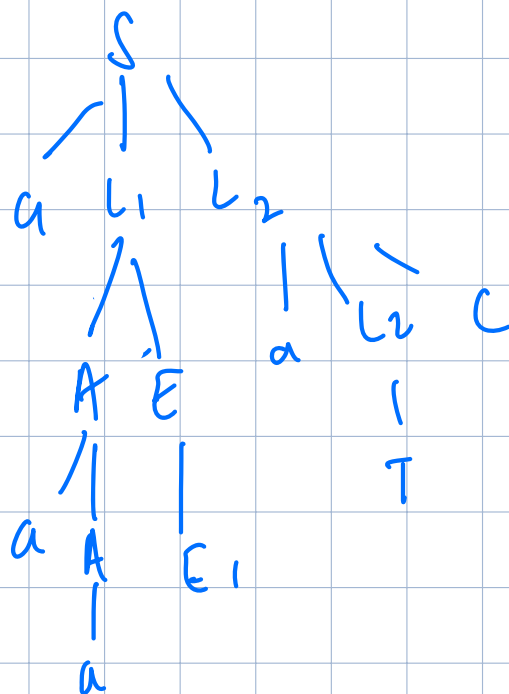
$$L_1 \rightarrow AE$$

$$A \rightarrow aA \mid a$$

$$L_2 \rightarrow aL_2c \mid T_1$$

$$T_1 \rightarrow bT_1c \mid bT_1 \mid b$$

aa bb c



$L \rightarrow aLc \mid L_1 \mid L_2$

$L_1 \rightarrow aL_1c \mid aL_4$

$L_2 \rightarrow bL_2c \mid L_3$

$L_3 \rightarrow bL_3c \mid b$

$L_4 \rightarrow bL_4c \mid L_3 \mid \epsilon$